Geometric applications of Linear Algebra

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Matrix

Transformations

$\mathcal{T} : \mathbb{R}^m \to \mathbb{R}^n$

$\mathcal{T}(\mathcal{u}) = \mathcal{v}$

$\forall\ \mathcal{u} \in \mathbb{R}^m, \forall \mathcal{v} \in \mathbb{R}^n$

Non-Linear Transformation

$\mathcal{T} : \mathbb{R}^3 \to \mathbb{R}^2$

$\mathcal{T}(x, y, z) = (2xy, z)$

$\mathcal{T}(1,4,2) = (8,2)$

Linear Transformation

$\mathcal{T} : \mathbb{R}^3 \to \mathbb{R}^2$

$\mathcal{T}(x, y, z) = (2x, y - z)$

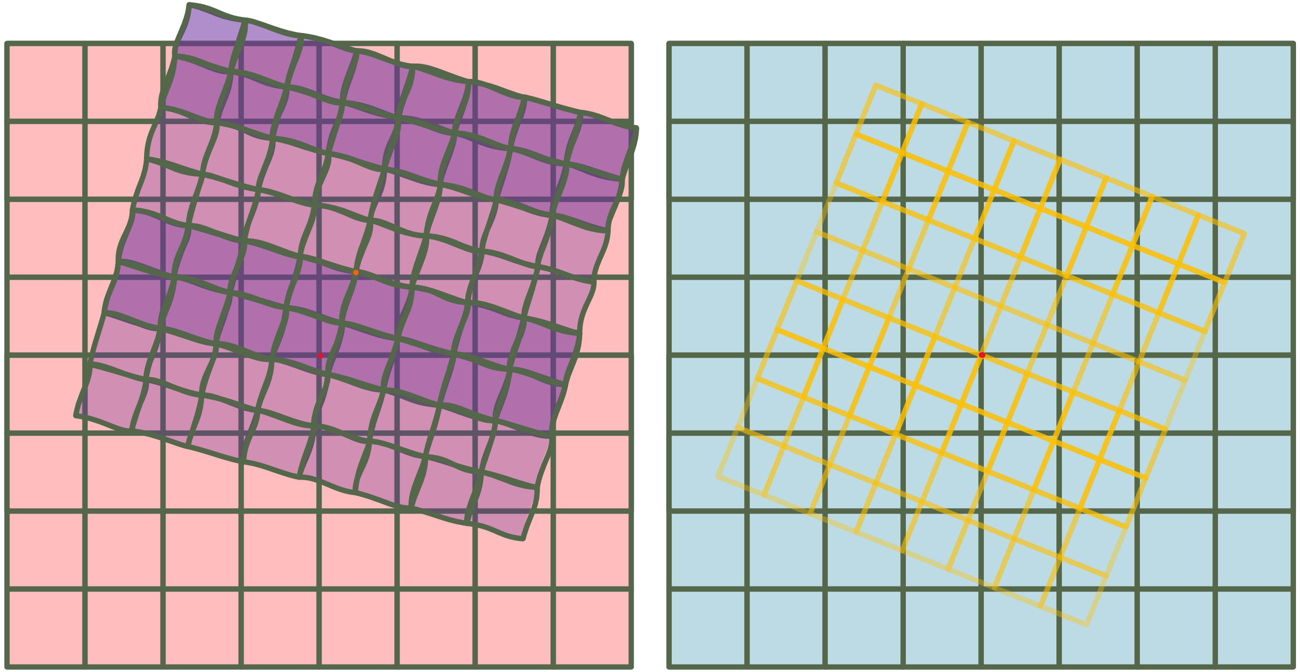
$\mathcal{T}(1,4,2) = (2,2)$

Linear Transformation

Transformation is linear if this statement is true:

$$\mathcal{T}(cu + v) = \mathcal{T}(cu) + \mathcal{T}(v) = c\,\mathcal{T}(u) + \mathcal{T}(v)$$

Prove linearity or non-linearity of given examples



Matrix as a transformation

$$\mathcal{T}(u) = \begin{bmatrix} a & \cdots & b \\ \vdots & \ddots & \vdots \\ c & \cdots & d \end{bmatrix} u$$

Additivity

$$\mathcal{M}(u+v) = \mathcal{M}(u) + \mathcal{M}(v)$$

scaling

$$\mathcal{M}(cu) = c\mathcal{M}(u)$$

Deeper understanding of a matrix

Recall that multiplying a matrix into a vector was described with this pattern:

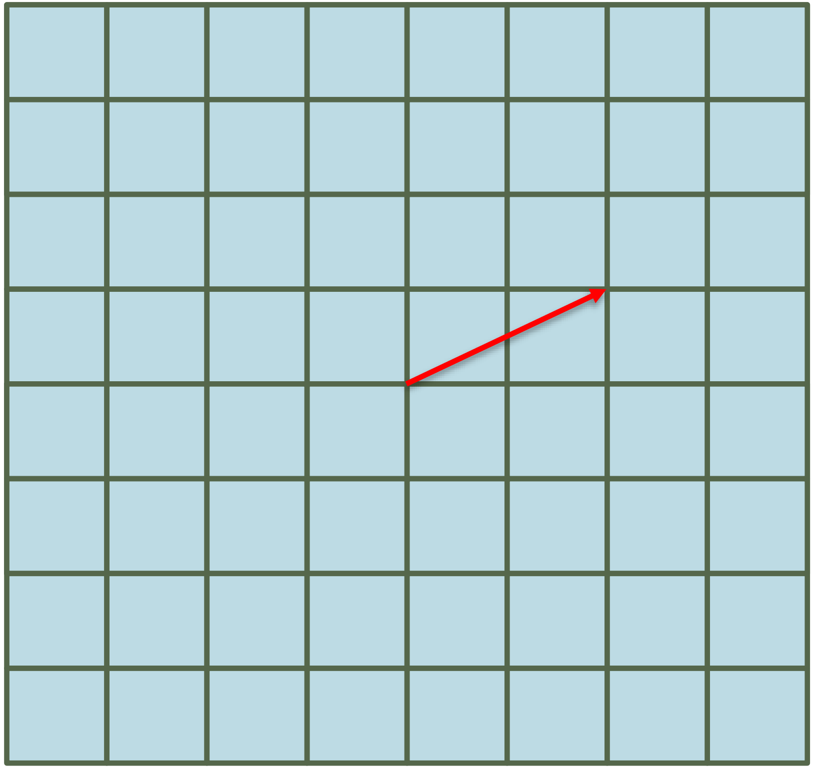
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

here is why:

$$\begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

i j

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = x \begin{bmatrix} a \\ b \end{bmatrix} + y \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

TABLE 1 Reflections

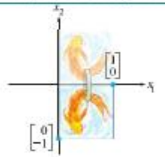
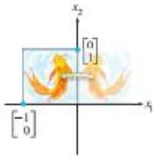
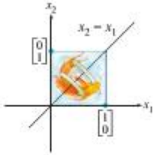
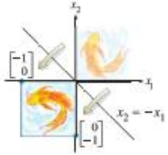
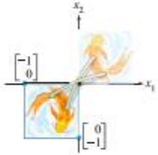
Transformation	Image of the Unit Square	Standard Matrix
Reflection through the x_1-axis		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Reflection through the x_2-axis		$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
Reflection through the line $x_2 = x_1$		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Reflection through the line $x_2 = -x_1$		$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$
Reflection through the origin		$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

TABLE 2 Contractions and Expansions

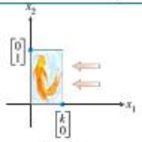
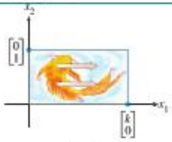
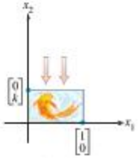
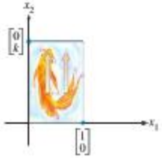
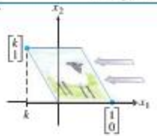
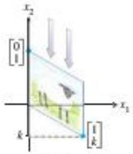
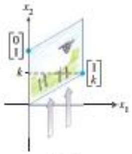
Transformation	Image of the Unit Square		Standard Matrix
Horizontal contraction and expansion	 $0 < k < 1$	 $k > 1$	$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$
Vertical contraction and expansion	 $0 < k < 1$	 $k > 1$	$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$

TABLE 3 Shears

Transformation	Image of the Unit Square		Standard Matrix
Horizontal shear	 $k < 0$	 $k > 0$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$
Vertical shear	 $k < 0$	 $k > 0$	$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$

Composition of 2 matrices

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{array}{cc} \begin{bmatrix} m & o \\ n & p \end{bmatrix} \\ \begin{array}{cc} i \quad\quad & j \end{array} \end{array}$$

$$= \left[m \begin{bmatrix} a \\ b \end{bmatrix} + n \begin{bmatrix} c \\ d \end{bmatrix} \quad o \begin{bmatrix} a \\ b \end{bmatrix} + p \begin{bmatrix} c \\ d \end{bmatrix} \right] =$$

$$\begin{bmatrix} ma + cn & oa + cp \\ mb + ob & nd + pd \end{bmatrix}$$

Dimensional transformations

$$\begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}_{3x2} : \mathbb{R}^2 \to \mathbb{R}^3$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix}_{3x1} : \mathbb{R}^1 \to \mathbb{R}^3$$

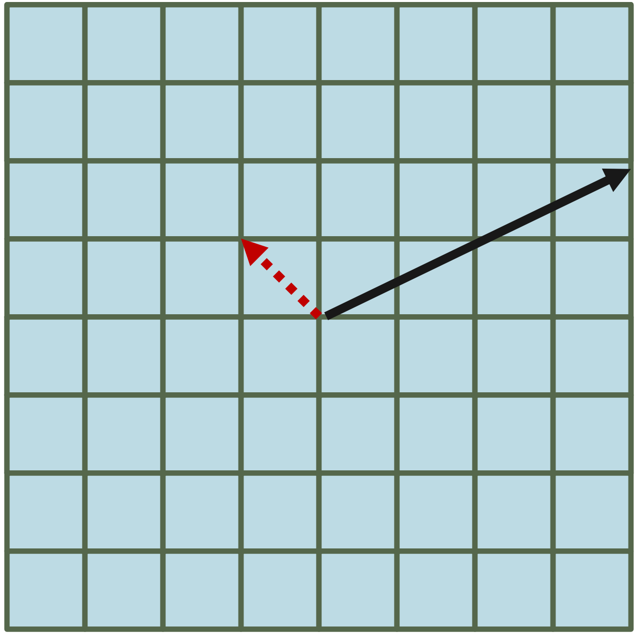
$$\begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix}_{2x3} : \mathbb{R}^3 \to \mathbb{R}^2$$

$$\begin{bmatrix} a & b & c \end{bmatrix}_{1x3} : \mathbb{R}^3 \to \mathbb{R}^1$$

Linear equations

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}$$



$$\mathrm{Ax} = \mathrm{b}$$

$$\begin{bmatrix} -3 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$\mathrm{x} = \mathrm{A}^{-1}\mathrm{b}$$

$$\mathrm{x} = \frac{-1}{14} \begin{bmatrix} 4 & -1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Eigenvalues/Eigenvectors

$$AX = \Delta X$$

$$(A-\Delta)X=0$$

$$X \mathrel{!}= 0$$

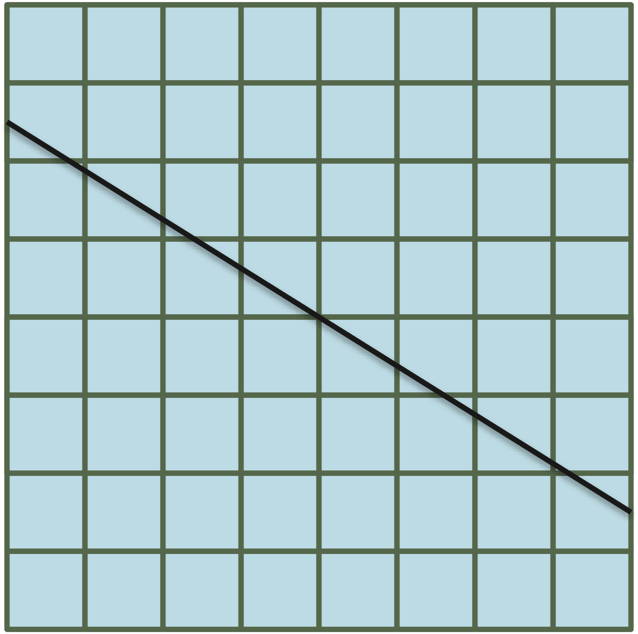
$$\det(A-\Delta I)=0$$

Mathmatical statements above indicate that there is a vector X that is immune to matrix transformation A. Only one attribute of X is affected by A transformation and that is it’s length, Δ.

Rank

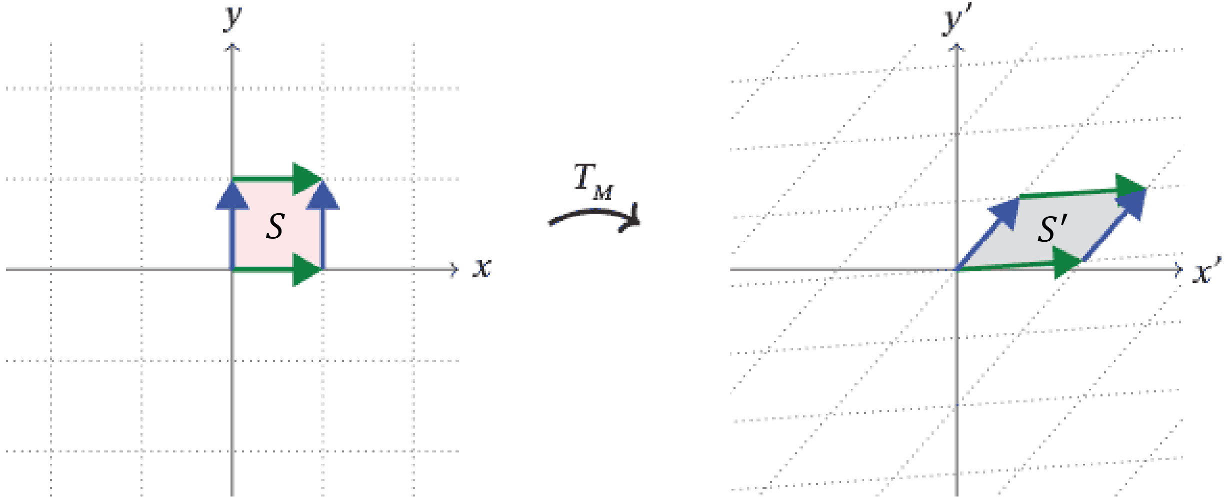
One common definition is that rank is the number of basis vectors in column/row space.

Away from the formal definition, rank is the number of dimensions after matrix transformation



02
Determinant

Determinant in Linear Transformation



Theorem: Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation determined by a 2×2 matrix A. If S is a parallelogram in $\mathbb{R}^2$, then

$$\{\text{area of T(S)}\} = |\text{detA}|.\{\text{area of S}\}$$

Proof: $A = [a_1 \quad a_2] \implies$ **Matrix of Linear Transformation**

$S = \{s_1 b_1 + s_2 b_2 : 0 \le s_1 \le 1, 0 \le s_2 \le 1\} \implies$ **A Parallelogram <u>at the origin</u> in** $\mathbb{R}^2$

$$T(S) = T(s_1 b_1 + s_2 b_2) = s_1 T(b_1) + s_2 T(b_2) = s_1 A b_1 + s_2 A b_2$$

$\xrightarrow{0 \le s_1 \le 1 \text{ and } 0 \le s_2 \le 1}$ $T(S)$ **is a Parallelogram determined by the columns of matrix** $[Ab_1 \quad Ab_2]$

$[Ab_1 \quad Ab_2] \xrightarrow{B = [b_1 \quad b_2]} [Ab_1 \quad Ab_2] = \text{AB} \implies \{\text{area of T(S)}\} = |\text{detAB}| = |\text{detA}|.|\text{detB}| = |\text{detA}|.\{\text{area of S}\}$

Proof (Continued):

$p + S$ An arbitrary Parallelogram

A vector

A parallelogram at the origin

$$
\begin{aligned}
\{\text{area of } \mathrm{T}(\mathrm{p}+\mathrm{S})\} &= \{\text{area of } \mathrm{T}(\mathrm{S})+\mathrm{T}(\mathrm{p})\} \\
&= \{\text{area of } \mathrm{T}(\mathrm{S})\} && \textbf{Translation} \\
&= |\det A|.\{\text{area of } \mathrm{S}\} \\
&= |\det A|.\{\text{area of } (\mathrm{p}+\mathrm{S})\} && \textbf{Translation}
\end{aligned}
$$

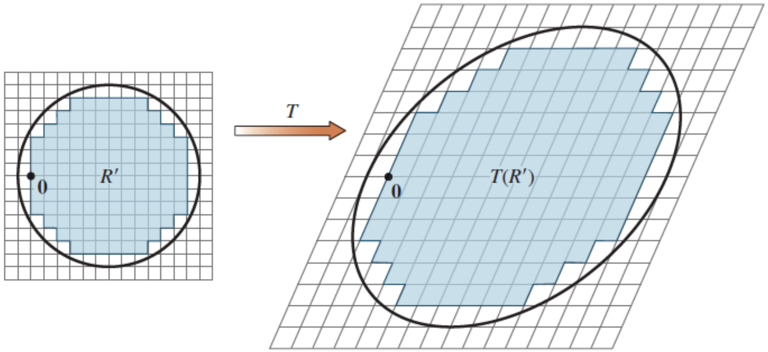
So this shows that theorem holds for all parallelograms in $\mathbb{R}^2$. ∎

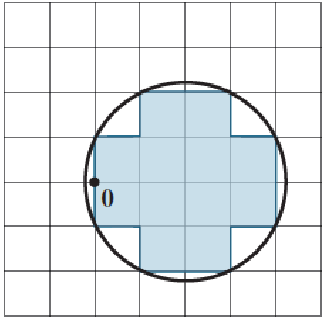
Theorem: Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation determined by a 3×3 matrix A. If S is a parallelepiped in $\mathbb{R}^3$, then

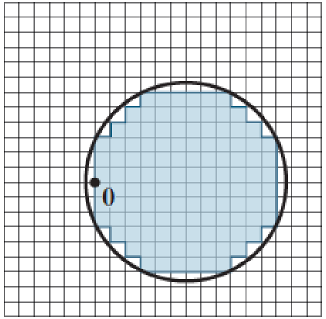
$$\{\text{volume of T(S)}\} = |\text{detA}|.\{\text{volume of S}\}$$

Proof:

Similar to Previous Theorem

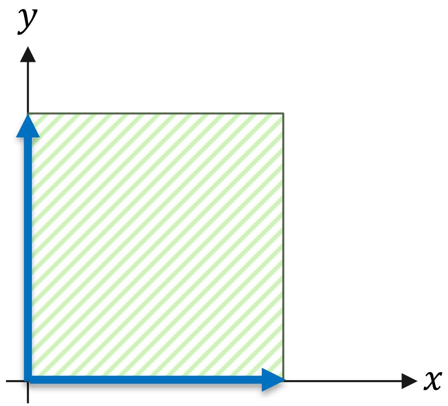


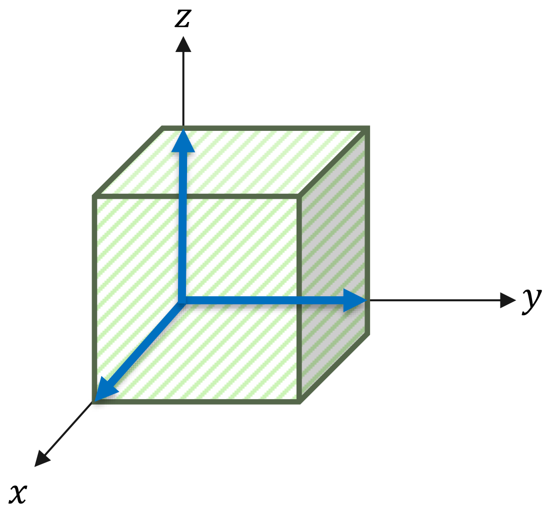




The conclusions of Previous hold whenever S is a region in $\mathbb{R}^2$ with finite area or a region in $\mathbb{R}^3$ with finite volume.

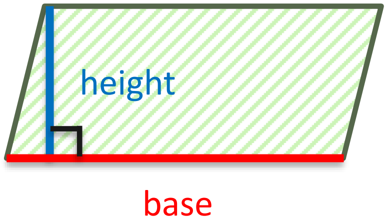
Determinant as Area and Volume





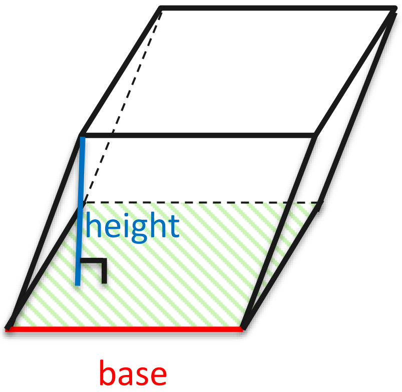
For a parallelogram, we have the familiar formula:

$$\begin{pmatrix}\text{area of}\\ \text{parallegram}\end{pmatrix} = (\text{length of base}) \times (\text{height})$$



We have a similar formula for the volume of a parallelepiped:

$$\begin{pmatrix}\text{volume of}\\ \text{parallelepiped}\end{pmatrix} = (\text{area of base}) \times (\text{height})$$

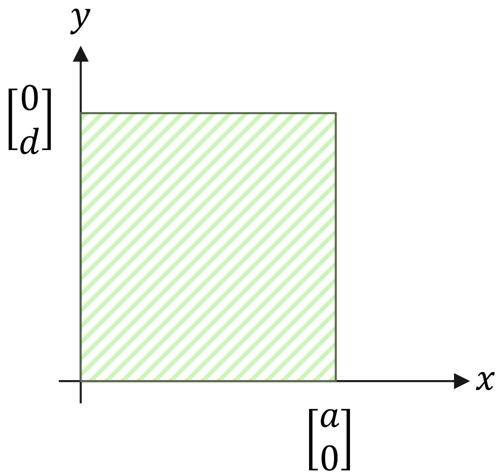


Theorem: if A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$.

Proof:

The theorem is obviously true for any 2×2 diagonal matrix:

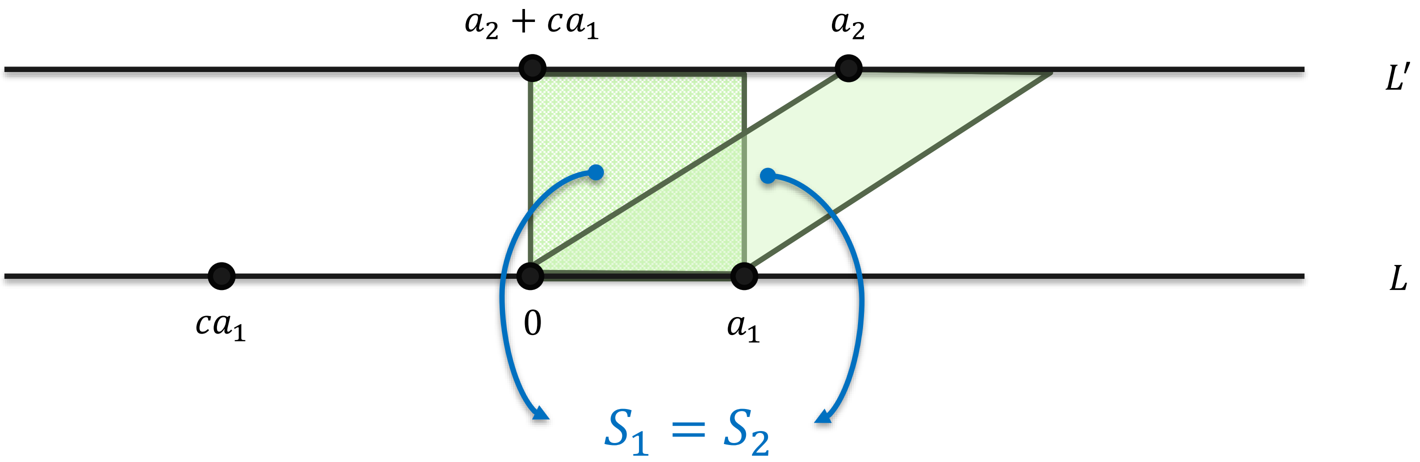
$$\begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} = |ad| = \begin{Bmatrix} \text{area of} \\ \text{rectangle} \end{Bmatrix}$$



Lemma: Let a_1 and a_2 be nonzero vectors. Then for any scalar c, the area of the parallelogram determined by a_1 and a_2 equals the area of the parallelogram determined by a_1 and $a_2 + ca_1$.

Proof:

Suppose that a_2 is not a multiple of a_1:



Theorem: if A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.

Proof:

The theorem is obviously true for any 3×3 diagonal matrix:

$$\begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = |abc| = \begin{Bmatrix} \textit{volume of} \\ \textit{cube} \end{Bmatrix}$$